

Solar Calculator — Using HTML, CSS, and JavaScript

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Internship: Direct Watts Interns

Solar Calculator

Built by Shivanshu Pokhriyal — Direct Watts Interns ⚡

Monthly Electricity Bill (₹)

e.g., 2500

Average Sunlight Hours per Day

e.g., 5

Cost per Watt (₹)

55

Panel Wattage (W)

350

Electricity Rate (₹/unit)

8

Calculate

Reset

Results

Estimated System Size	— kW
Approx. Number of Panels	—
Total System Cost	— ₹

A — kW system can save around ₹— /month.

1. Project Overview: -

The Solar Calculator is a web-based tool that helps users estimate the size, cost, and savings of a solar power system based on their monthly electricity usage. Users can input their electricity bill, average sunlight hours, cost per watt, panel wattage, and electricity rate to calculate the required solar system capacity, number of panels, and total cost.

2. HTML Structure: -

The HTML file provides the structure for the entire web page. It includes sections for header, input form, and results display. Each input is wrapped in a <label> for accessibility and connected to JavaScript using unique IDs.

Key Sections:

- Header— Displays title and Developer info.

```
1 <!DOCTYPE html>
2 <html lang="hi">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Solar Calculator - Direct Watts Interns</title>
7   <link rel="stylesheet" href="style.css">
8   <script src="https://kit.fontawesome.com/a2e0e6a91f.js" crossorigin="anonymous"></script>
9 </head>
10 <body>
11   <main class="container">
12     <header>
13       <h1><i class="fa-solid fa-solar-panel"></i> Solar Calculator</h1>
14       <p class="subtitle">Built by <strong>Shivanshu Pokhriyal</strong> - Direct Watts Interns ⚡</p>
15     </header>
```

- Input Section — Accepts user data including Monthly Bill, Sunlight Hours, Cost per Watt, Panel Wattage, and Electricity Rate.
- Action Buttons — ‘Calculate’ and ‘Reset’ buttons for functionality.

```
16 <section class="card inputs">
17   <label>
18     <i class="fa-solid fa-bolt"></i> Monthly Electricity Bill (₹)
19     <input id="monthlyBill" type="number" placeholder="e.g., 2500">
20   </label>
21
22   <label>
23     <i class="fa-solid fa-sun"></i> Average Sunlight Hours per Day
24     <input id="sunHours" type="number" placeholder="e.g., 5">
25   </label>
26
27   <label>
28     <i class="fa-solid fa-coins"></i> Cost per Watt (₹)
29     <input id="costPerWatt" type="number" value="55">
30   </label>
31
32   <label>
33     <i class="fa-solid fa-solar-panel"></i> Panel Wattage (W)
34     <input id="panelWatt" type="number" value="350">
35   </label>
36
37   <label>
38     <i class="fa-solid fa-battery-full"></i> Electricity Rate (₹/unit)
39     <input id="unitRate" type="number" value="8">
40   </label>
41
42   <div class="actions">
43     <button id="calculateBtn"><i class="fa-solid fa-calculator"></i> Calculate</button>
44     <button id="resetBtn" class="muted"><i class="fa-solid fa-rotate-right"></i> Reset</button>
45   </div>
46 </section>
```

- Results Section — Displays calculated system size, number of panels, total cost, and estimated monthly savings.

```
49 <section class="card results">
50   <h2><i class="fa-solid fa-chart-line"></i> Results</h2>
51   <div class="result-row"><span>Estimated System Size</span><strong id="systemSize">- KW</strong></div>
52   <div class="result-row"><span>Approx. Number of Panels</span><strong id="panels">-</strong></div>
53   <div class="result-row"><span>Total System Cost</span><strong id="totalCost">- ₹</strong></div>
54   <p id="savings" class="savings">A <span id="roundedKW">-</span> kW system can save around ₹<span id="saveAmt">-</span>
55   <span> /month.</p>
56 </section>
57
58 <script src="script.js"></script>
59 </body>
60 </html>
61
```

3. CSS Styling: -

The CSS (style.css) provides a modern and responsive design using gradients, transparency, and smooth animations. It applies a glass morphism effect for card backgrounds, rounded corners, and soft shadows for a clean and futuristic look.

Styling Features:

- Background — Gradient background with shades of blue-green.
- Font — 'Poppins' imported from Google Fonts for clean typography.
- Inputs and Buttons — Styled with padding, rounded corners, and hover effects.
- Responsive Design — Media queries ensure compatibility across devices.

```
# style.css > @keyframes fadeIn
1  @import url('https://fonts.googleapis.com/css2?family=Poppins:wght@400;600;700&display=swap');
2
3  :root {
4    --bg1: #0f2027;
5    --bg2: #203a43;
6    --bg3: #2c5364;
7    --card-bg: rgba(255, 255, 255, 0.15);
8    --border: rgba(255, 255, 255, 0.3);
9    --text: #ffffff;
10   --accent: #22d3ee;
11   --shadow: 0 8px 32px rgba(0, 0, 0, 0.25);
12 }
13
14 * {
15   box-sizing: border-box;
16 }
17
18 body {
19   margin: 0;
20   font-family: 'Poppins', sans-serif;
21   color: var(--text);
22   background: linear-gradient(135deg, var(--bg1), var(--bg2), var(--bg3));
23   min-height: 100vh;
24   display: flex;
25   align-items: center;
26   justify-content: center;
27 }
28
29 @keyframes fadeIn {
30   from { opacity: 0; transform: translateY(15px); }
31   to { opacity: 1; transform: translateY(0); }
32 }
33
34 @media (max-width: 480px) {
35   header h1 { font-size: 1.4rem; }
36   button { font-size: 0.9rem; }
37 }
38
39
40
41
42
43
44
45
```

4. JavaScript Functionality: -

The JavaScript (script.js) controls all the calculations and interactivity. It collects user inputs, validates them, performs mathematical calculations, and displays results dynamically without reloading the page.

Input Fields:

- Monthly Electricity Bill (₹)
- Average Sunlight Hours per Day
- Cost per Watt (₹)
- Panel Wattage (W) — lets user define solar panel capacity (default 350W)
- Electricity Rate (₹/unit) — allows users to enter their actual electricity rate (default ₹8/unit)

```

JS scripts > calculate
1  const monthlyBillEl = document.getElementById('monthlyBill');
2  const sunHoursEl = document.getElementById('sunHours');
3  const costPerWattEl = document.getElementById('costPerWatt');
4  const panelWattEl = document.getElementById('panelWatt');
5  const unitRateEl = document.getElementById('unitRate');
6
7  const calcBtn = document.getElementById('calculateBtn');
8  const resetBtn = document.getElementById('resetBtn');
9
10 const systemSizeEl = document.getElementById('systemSize');
11 const panelsEl = document.getElementById('panels');
12 const totalCostEl = document.getElementById('totalCost');
13 const roundedKWEl = document.getElementById('roundedKW');
14 const saveAmtEl = document.getElementById('saveAmt');
15

```

Main Functions:

- format Rupee () — Formats numbers into Indian currency format.
- calculate () — Fetches user input, validates data, performs all solar calculations, and updates the result section dynamically.
- resetBtn — Clears all fields and resets the display to default.

```

16 function formatRupee(num) {
17   return Number(num).toLocaleString('en-IN', { maximumFractionDigits: 0 });
18 }
19
20 function calculate() {
21   const bill = parseFloat(monthlyBillEl.value);
22   const sunHours = parseFloat(sunHoursEl.value);
23   const costPerWatt = parseFloat(costPerWattEl.value);
24   const panelWatt = parseFloat(panelWattEl.value);
25   const unitRate = parseFloat(unitRateEl.value);
26
27   if (!bill || !sunHours || !panelWatt || !unitRate || bill <= 0 || sunHours <= 0) {
28     alert('Please enter valid electricity bill and sun hours!');
29     return;
30   }
31
32   const monthlyUnits = bill / unitRate;
33   const dailyUsage = monthlyUnits / 30;
34   const systemKW = dailyUsage / sunHours;
35   const systemKWdisplay = Math.round(systemKW * 100) / 100;
36
37   const totalWatts = systemKW * 1000;
38   const panelsNeeded = Math.ceil(totalWatts / panelWatt);
39   const totalCost = Math.round(totalWatts * costPerWatt);
40
41   const monthlySavingEstimate = Math.round(bill);
42
43   systemSizeEl.textContent = `${systemKWdisplay} kW`;
44   panelsEl.textContent = panelsNeeded;
45   totalCostEl.textContent = `₹ ${formatRupee(totalCost)}`;
46   roundedKWEl.textContent = systemKWdisplay;
47   saveAmtEl.textContent = formatRupee(monthlySavingEstimate);
48
49   document.querySelector('.results').classList.add('highlight');
50   setTimeout(() => document.querySelector('.results').classList.remove('highlight'), 800);
51 }
52
53 calcBtn.addEventListener('click', calculate);
54
55 resetBtn.addEventListener('click', () => {
56   monthlyBillEl.value = '';
57   const costPerWattEl: HTMLElement | null
58   costPerWattEl.value = '55';
59   panelWattEl.value = '350';
60   unitRateEl.value = '8';
61   systemSizeEl.textContent = '- kW';
62   panelsEl.textContent = '-';
63   totalCostEl.textContent = '- ₹';
64   roundedKWEl.textContent = '-';
65   saveAmtEl.textContent = '-';
66 });
67

```

Calculation Steps:

1. Get user inputs for all fields including new wattage and rate values.
2. Validate values to ensure all inputs are filled correctly.
3. Calculate monthly units = monthly bill / electricity rate.
4. Estimate required solar system size based on average sunlight hours.
5. Determine number of panels = total watts / panel wattage.
6. Calculate total cost = total watts × cost per watt.
7. Display all calculated results dynamically on the webpage.